

ORIGINAL CONTRIBUTION

Susceptibility of *Dendroctonus simplex* to Hypocreales fungi: towards the development of a biological control strategyN. Srei¹, R. Lavallée² & C. Guertin¹¹ INRS-Institut Armand-Frappier, Laval, QC, Canada² Laurentian Forestry Centre, Natural Resources Canada, Canadian Forest Service, Québec, QC, Canada**Keywords**assisted autodissemination, biological control, *Dendroctonus simplex*, Hypocreales fungi**Correspondence**Claude Guertin (corresponding author), INRS-Institut Armand-Frappier, Laval, QC, Canada H7V 1B7. E-mail: claud.guertin@iaf.inrs.ca

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Abstract

The use of Hypocreales fungi against bark beetles is an attractive alternative strategy for biological control in forestry. However, fungal inoculation methods remain a major challenge due to the cryptic behaviour of these beetles. Experiments were conducted to evaluate the susceptibility of adult eastern larch beetles, *Dendroctonus simplex*, to different isolates of Hypocreales fungi and to assess the efficiency of direct inoculation of fungal conidia using an assisted autodissemination device. Among the nine fungal isolates tested, *Beauveria bassiana* INRS-242 was selected as the utmost isolate causing a mortality rate of 64.4% of *D. simplex* adults within 10 days and having a high conidial viability of 81.7%. Using an assisted autodissemination device, a fungal-coated pouch was a promising contamination chamber. Once it had fallen onto the fungal-coated pouch, each *D. simplex* adult loaded an average lethal dose of 2.1×10^9 conidia, which induced a mortality rate of 98.0%. Our results demonstrated the fungal pathogenicity of *B. bassiana* INRS-242 against the eastern larch beetles and pointed out the potential of the assisted autodissemination device as an effective method of conidia inoculation. These findings represent the first step to develop a biological control strategy against *D. simplex* populations.

Introduction

The eastern larch beetle, *Dendroctonus simplex* LeConte (Coleoptera: Scolytinae), is an indigenous insect that may cause serious damage in coniferous forests in North America. The insect geographical distribution follows its host tree, the tamarack (*Larix laricina* (Du Roi) K. Koch), from Newfoundland to British Columbia and the northern United States to Alaska (Langor and Raske 1987a,b; Seybold et al. 2002). It is considered as a secondary pest attacking mainly stressed or recently dead trees during the endemic phase. However, *D. simplex* can attack healthy trees during the epidemic phase, especially when climate conditions promote an outbreak of insect populations (Benoit and Blais 1984; Seybold et al. 2002; Bentz et al. 2010). Infestations of *D. simplex* were first reported in 1883 in North America, in 1926 in the

province of Quebec and in 1946 in western Canada. During the late 1970s and early 1980s, around 1.4 million m³ of tamarack stands were destroyed by *D. simplex* (Seybold et al. 2002). In Alaska, *D. simplex* infested around 3.3 million m³ of tamarack stands. In 1984, during a survey conducted in southern Quebec by Benoit and Blais (1984), over 50% of the visited tamarack stands were attacked by *D. simplex*. Other important infestations were concomitantly observed in northern Minnesota (Seybold et al. 2002; Minnesota Department of Natural Resources - Division of Forestry 2014), in Ontario (Ontario Ministry of Natural Resources 2012), in Manitoba (Manitoba Conservation Forestry Branch 2014) and in Alberta (Mckee and Aukema 2015). Despite no recent cumulative data on tamarack mortalities in Quebec province, during the last 5 years, we observed damage associated with *D. simplex* in different regions, namely in the Parc

des Laurentides, north and south of the St. Lawrence River near the Quebec City area, south of Montreal and in the Eastern Townships.

A few directly targeted management strategies at the eastern larch beetles were evaluated in the past. However, some strategies based on pesticide treatments of trunks could reduce attacks of *D. simplex* (Seybold et al. 2002). As treatments with chemical pesticides are known harmful to the environment and human (Sezen et al. 2004; Muratoglu et al. 2011; Kaya and Vega 2012), researches on developing more effective and safe biological control agents are warranted. Like many other bark beetle species, the adults and larvae of *D. simplex* feed on phloem during their development and maturation under the bark. In this context, the use of Hypocreales fungal species, such as *Beauveria bassiana* (Bals.) Vuill., *Metarhizium anisopliae* (Metschn.) Sorokin and *Lecanicillium* spp. W. Gams & Zare, was considered to be suitable as biological control agents against bark beetle populations (Kreutz et al. 2004; Wegensteiner 2004; Tanyeli et al. 2010; Kocacevik et al. 2015, 2016). Even though Hypocreales fungi invade their hosts directly through the cuticle of insects, fungal inoculation methods remain a major challenge in the case of most bark beetle species because of their cryptic behaviour. The commonly used methods are either by immersion or by spraying, and conidia are generally formulated in suspension or powder (Kreutz et al. 2004; Hajek et al. 2006; Liu and Bauer 2006). These methods are suitable to control insects of the foliage or surface. Additionally, viability and insecticidal activities of suspended conidia are reduced when exposed to environmental conditions (Behel 2006). In this context, much research has focused on direct inoculation of conidia using an assisted autodissemination device (Baverstock et al. 2010). This device requires live trapping of the pest species in large numbers and their contamination with fungal conidia to cause fungus dissemination from infected insects to their conspecifics (Wegensteiner et al. 2015). In the past, this strategy has been successfully used against different species of Coleoptera (Vega et al. 1995, 2007; Dowd and Vega 2003; Shimazu 2004; Lyons et al. 2012), Diptera (Maniania et al. 2011) and Lepidoptera (Pell et al. 1993). However, little is known about the efficiency of the assisted fungal autodissemination device to control bark beetles.

The aim of this study was to evaluate the susceptibility of *D. simplex* adults to Hypocreales fungi and select the isolate causing the highest mortality. Evaluation of the pathogenicity degree of the selected isolate and the efficiency of the direct inoculation

method on adult *D. simplex* using an assisted fungal autodissemination device were also investigated. These findings represent the first step to develop a biological control strategy against *D. simplex* populations.

Materials and Methods

Insects

The adults of *D. simplex* used in this study originated from infested tamarack private stands and government plantations located, respectively, at Sainte-Catherine-de-la-Jacques-Cartier (QC, Canada; lat. 46.830810 and long. -71.571361) and Cleveland arboretum (St-Claude, QC, Canada; lat. 45.6809 and long. -71.9969). The infested trees were randomly selected and felled trees were then cut into 50-cm log sections for transportation with the permission of the stand owners and the Quebec ministère des Forêts, de la Faune et des Parcs authorization. Under laboratory conditions, the bark was gently peeled off and adults were carefully collected with sterilized tweezers and then individually placed in a Petri dish (50 × 9 mm, BD Falcon™, USA), containing a piece of tamarack twig as feeding substrate and a filter paper (3.2 cm, Whatman International Ltd., Maidstone, UK). Each filter paper was moistened by adding 120 µl of sterile distilled water. The Petri dishes were set in a vertical position in a rearing incubator (SANYO MLR-350, Osaka, Japan) during 3 weeks at 20°C, 70% relative humidity and complete darkness. Only vigorous adults were used for the experiments.

Fungal production

Isolates used in this study came from the fungal collection of the INRS-Institut Armand-Frappier (Laval, QC, Canada). Nine isolates of Hypocreales fungi from three species, including *B. bassiana* (Bals.) Vuill. (six isolates), *Metarhizium anisopliae* (Metschn.) Sorokin (one isolate) and *Lecanicillium attenuatum* Zare & W. Gams (two isolates), were selected for the screening tests (Table S1). They were isolated from diverse substrates or hosts from different geographical origins. The isolates named NRCAN-Val-Bb2, NRCAN-TN6-Bb, NRCAN-PN069A and NRCAN-PN077C were graciously provided by Robert Lavallée's laboratory.

Fungal production was carried out in two steps by following the previous work of Srei (2011) with some modifications. First, selected isolates were grown in yeast extract potato dextrose, YPD (Alpha Bioscience Inc., Baltimore, MD, USA) at 25°C for 4 days in a rotary shaker (150 rpm). Then, mass production was

achieved to produce conidial suspensions for all the downstream experiments. Each isolate was cultured in plastic fungal spawn bags with a 0.22 μm filter (Fungi Perfecti, Olympia, WA, USA), which ensured air exchange during fungal growth. Each bag contained 100 g of pearled barley (Nith River Milling Inc., Linwood, ON, Canada) and 100 ml of distilled water. After sterilization at 121°C for 20 min, the bags were left cooling overnight at room temperature. For each isolate, the culture bags were inoculated with 10 ml of fungal suspension previously produced in YPD. All bags were then sealed and placed in a growth chamber, at 25°C in darkness for 14 days. Clumps of culture medium were manually broken at day 7 to provide sufficient aeration in the medium. Each isolate was produced in triplicate. Conidia of each isolate were harvested by adding 200 ml of sterile distilled water. Conidial suspensions were then filtered through three layers of sterile cheesecloth to remove mycelia and barley residuals. The concentration of each conidial suspension was quantified under a microscope with a hemocytometer (Bright-Line Improved Neubauer; Hausser Scientific, Horsham, PA, USA). All conidial suspensions were kept at 4°C until use.

Conidial viability

Two days before each bioassay, the conidial viability of each isolate was evaluated by counting the germinated and non-germinated conidia using the technique of Meyling and Pell (2006) with some modifications. About 20 μl of 1×10^8 conidia ml^{-1} was spread on Sabouraud dextrose agar, SDA (Difco Laboratories, Mannheim, Germany) plate, followed by incubation at 20°C in darkness for 36 h. In three randomly selected zones, 100 conidia (including germinated and non-germinated conidia) per zone were counted. Conidia of isolates of *Metarhizium anisopliae* and *Lecanicillium attenuatum* are considered viable if germ tube lengths are twice the conidial diameter. Particularly for *B. bassiana* isolates, swelling of germinating conidia was used to indicate viability (Inglis et al. 2012). For each isolate, three plates were incubated and the average of germination percentage of counted conidia was calculated.

Isolate screening

To select the most virulent isolate against *D. simplex* adults, screening tests were performed using the nine isolates of Hypocreales fungi mentioned above (table S1) according to Butt et al. (1994) with some

modifications. For each isolate, groups of 20 adults were inoculated by immersion for 10 s in 10 ml of 1×10^8 conidia ml^{-1} , and insects were individually transferred into sterile Petri dishes (50 $\times$ 9 mm) containing a sterile wet filter paper, and a piece of tamarack twig (1.5 cm). They were then placed in a growth chamber at 20°C, 70% relative humidity and in darkness. Mortality was recorded daily for 10 days. Mycosis was confirmed by returning the dead eastern larch beetle in a growth chamber for another 14 days. The appearance of white muscardine on cadavers indicated mycosis. Control insects consisted of 20 adults immersed in 10 ml of sterile distilled water, and they were maintained under controlled incubation conditions as described above. All experiments were repeated three times, each time with one of the conidia batch from mass production. Cumulative percentage of mortality data from the isolate screening tests were corrected for natural mortality using Abbott's formula (Abbott 1925). The most virulent isolate was identified based on the mean percentage of cumulative mortality after 10 days of inoculation, and the conidial viability of each isolate tested.

Fungal pathogenicity

Based on results from the screening tests on *D. simplex* adults, the isolate INRS-242 of *B. bassiana* was selected to evaluate its pathogenicity degree. Two variables were estimated: the lethal concentration (LC) and the median survival time (MST). From the stock of conidial suspensions of *B. bassiana* INRS-242, a serial dilution in sterile distilled water was performed to obtain six concentrations ranging from 1×10^6 to 1×10^{11} conidia ml^{-1} . For each concentration, groups of 28 insects were immersed for 10 s

Table 1 Pathogenicity of *Beauveria bassiana* INRS-242. The degree of pathogenicity of *B. bassiana* INRS-242 was evaluated by calculating the lethal concentration and median survival time of *D. simplex* adults after fungal infection. Three different lethal concentrations after 6 days of infection were estimated by probit analysis. The median survival time of *D. simplex* adults exposed to 1×10^9 conidia ml^{-1} was observed between 5 and 6 days (lifetest)

LC	Conidia ml^{-1}	95% confidence limits	
		Lower	Upper
LC ₁₀	2.1×10^7	9.3×10^6	3.9×10^7
LC ₅₀	7.4×10^8	4.8×10^8	1.1×10^9
LC ₉₀	2.6×10^{10}	1.4×10^{10}	6.0×10^{10}

in 10 ml of the conidial suspension and individually transferred into a Petri dish as previously described. Control insects consisting of 28 adults were also treated with 10 ml of sterile distilled water and subjected to the same incubation conditions. All concentrations were assayed in triplicate. Mycosis was confirmed by the presence of the white muscardine on cadavers.

Assisted autodissemination

An assisted fungal autodissemination device was used to evaluate the efficiency of direct inoculation of conidia on *D. simplex* adults. The device used was adapted from Lyons et al. (2012), and it consisted of two parts: a 12-funnel black Lindgren trap (Synergy Semiochemicals, Burnaby, BC, Canada) and a dissemination chamber with a fungal-coated pouch inoculated with *B. bassiana* INRS-242 isolate. Herein, two variables were measured: the number of conidia loaded per *D. simplex* adult and the efficacy of fungal infection following the passage of the beetles through the autodissemination device.

To determine whether conidia loaded per insect varied as a function of the insect's exposure time on the pouch, one insect at a time was dropped from the top of the autodissemination device and left walking on the fungal-coated pouch for 15, 30, 45, 60 and 120 s. A cohort of 15 *D. simplex* adults was used for each exposure time. To determine conidia load, each insect was individually transferred into a 600- μ l sterile tube containing 100 μ l of 0.1% Tween-80. Each tube was agitated twice using a Vortex Genie 2 for 2 min and left to decant for 2 min between agitations. The insect was then removed from the tube and discarded, and the conidial suspension was counted under a microscope with a hemocytometer.

To evaluate the efficiency of fungal infection following the passage through the assisted autodissemination device, 26 beetles were individually dropped directly from the top of the device, and left walking on the fungal-coated pouch for 1 min. Then, each insect was transferred into a sterile Petri dish containing a sterile wet filter paper, and a piece of tamarack twig. For the control group, insects were dropped from the top of the device with an uncoated pouch (without *B. bassiana* INRS-242) and also placed individually in a Petri dish. Dishes were incubated in a growth chamber at 20°C, 70% relative humidity and in darkness. Mortality and the presence of white muscardine were recorded every 24 h for 11 days. Experiments were repeated four times.

Statistical analysis

Data analyses were performed using the statistical software package SAS 9.2 (SAS Institute Inc., 2002). Percentage of corrected mortality and conidial viability were subjected to one-way analysis of variance, ANOVA, to show a significant overall difference in all isolates. Mean separation was then compared using the Duncan's multiple comparison procedure. Lethal concentration of *B. bassiana* INRS-242 for insect mortality was determined using a probit analysis (Finney 1971), whereas median survival time of insects after treatment with *B. bassiana* INRS-242 at a concentration of 1×10^9 conidia ml⁻¹ was estimated by a life-test analysis. Two-sample *t*-test was used to compare mycosis of insects after contact with the fungal-coated pouch and the uncoated pouch. Correlations between insect mortality and conidial viability (isolate screening), and between conidia loaded per insect and exposure time on the pouch were analysed using the Pearson regression. For all statistical tests, the level of rejection was set at $\alpha = 0.05$.

Results

Conidial viability

The viability of the conidia of all isolates was tested and expressed as the percentage of germinated conidia (Fig. 1). A significant difference in conidial

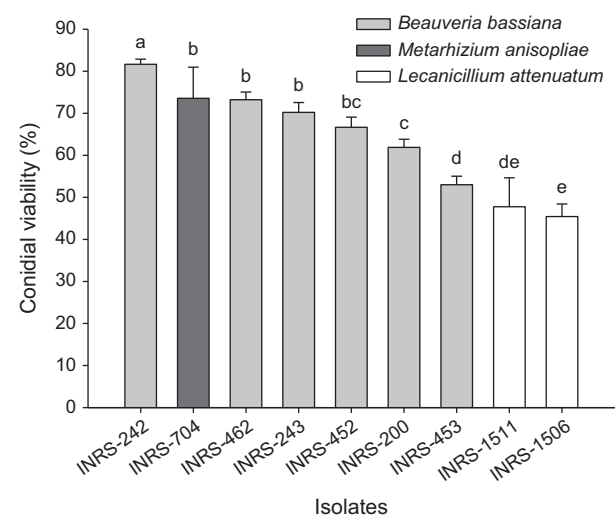


Fig. 1 Mean percentage ($\pm$ SD) of conidial viability of different fungal isolates used in screening tests. For each isolate, conidial viability was evaluated by counting the germinated conidia incubated at 20°C during 36 h. Different letters above bars indicate statistically significant (Duncan's test, $P < 0.05$).

viability was found between all isolates (ANOVA, $n = 27$, d.f. = 8, $F = 31.69$, $P < 0.0001$). Conidia from isolate INRS-242 of *B. bassiana* showed the highest germination with $81.7 \pm 1.2\%$. The lowest conidial viability was observed with isolates INRS-1506 and INRS-1511 of *L. attenuatum*, respectively, $45.4\% \pm 3.0\%$ and $47.5\% \pm 6.9\%$ of germinated conidia.

Isolate screening

The mortality rates of *D. simplex* adults exposed to different Hypocreales isolates at a concentration of 1×10^8 conidia ml^{-1} are shown in fig. 2. A significant difference was observed between tested isolates (ANOVA, $n = 27$, d.f. = 8, $F = 4.46$, $P = 0.004$). The highest mortality was obtained with the isolate INRS-242 of *B. bassiana* ($64.4\% \pm 10.8\%$), followed by the isolate INRS-704 of *M. anisopliae* ($55.7\% \pm 13.7\%$) and the isolate INRS-462 of *B. bassiana* ($43.5\% \pm 8.9\%$). However, no significant difference was observed between these last three isolates (Duncan's test, $P > 0.05$). The isolates INRS-453, INRS-200, INRS-452 and INRS-243 of *B. bassiana* induced mortality ranging from 25% to 30%. The lowest mortalities were observed with adults exposed to isolates INRS-1506 and INRS-1511 of *L. attenuatum*, with less than 25% of dead insects, and no significant difference was observed among them. A significant correlation was found between mortality of *D. simplex* adults exposed to the Hypocreales isolates and their conidial viabilities (Pearson, $n = 27$, $r^2 = 0.6$) (fig. 3). Based on results obtained from the screening tests and the conidial viability tests, isolate INRS-242 of *B. bassiana* was chosen for further experiments.

Fungal pathogenicity

The virulence of the INRS-242 isolate of *B. bassiana* was evaluated by calculating the lethal concentration (LC) and median survival time (MST) of *D. simplex* adults exposed to conidia. Mortality of insects exposed to the fungus increased with time and with conidial concentration (table 1). Using probit analysis, the concentration of conidia that killed 50% of the adult population (LC_{50}) was estimated to be about 7.4×10^8 conidia ml^{-1} , and the LC_{90} at about 2.6×10^{10} conidia ml^{-1} , both 6 days after the infection. Furthermore, the MST for *D. simplex* adults infected by the INRS-242 isolate at 1×10^9 conidia ml^{-1} was estimated to be between 5 and 6 days. Our results also indicated that there was a latency

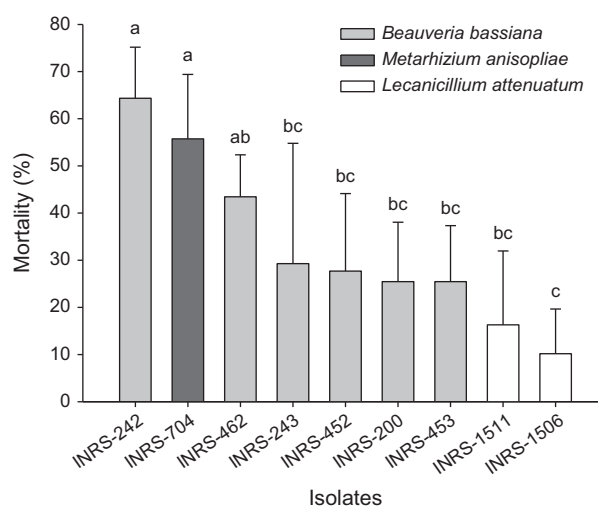


Fig. 2 Cumulative mortality of *D. simplex* adults ($\% \pm \text{SD}$) within 10 days after fungal exposition. *D. simplex* adults were treated with conidial suspension of fungal isolates at a concentration of 1×10^8 conidia ml^{-1} . Natural mortality was corrected based on Abbott's formula. Different letters above bars indicate statistically significant (Duncan's test, $P < 0.05$).

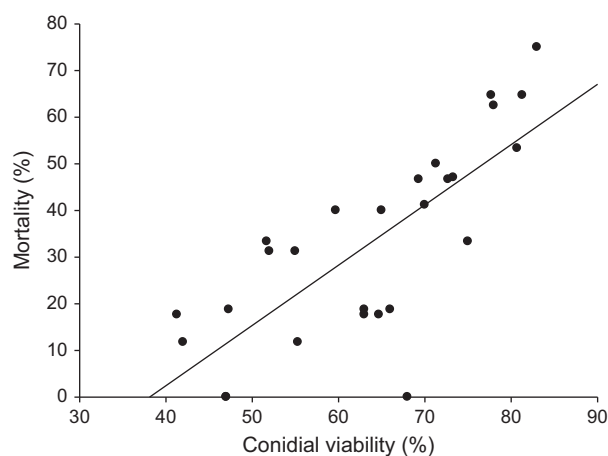


Fig. 3 Correlation between mortality of *D. simplex* exposed to fungi and the conidial viability associated with each isolate (Pearson test, $n = 75$; $r^2 = 0.6$; $P < 0.001$).

phase of 3 days before any mortality was observed in the insect population.

Assisted autodissemination

No correlation was found between the conidia loaded per insect and the duration of exposure on the fungal-coated pouch (Pearson test, $n = 75$, $r^2 = 0.08$) (fig. 4). In all cases, the average of conidia loaded was estimated at 2.1×10^9 conidia per beetle. The efficiency

of the autodissemination device using the INRS-242 isolate of *B. bassiana* was evaluated by recording mycosis on cadavers of *D. simplex* adults during 11 days after passing through the dissemination chamber. A significant difference was observed on mycosis symptoms associated with fungal-exposed and non-exposed insect cadavers (*t*-test, $n = 208$, $P < 0.0001$), with $98.1\% \pm 1.9\%$ on exposed adults ($n = 104$) and no mycosis symptoms on cadavers from the control group ($n = 104$).

Discussion

The goal of our study was to develop a biological control strategy against the eastern larch beetle. It is the first time that such an investigation was conducted to evaluate the susceptibility of these adult beetles to three entomopathogenic fungal species of the Hypocreales order. First, our results showed that *D. simplex* was very susceptible to three specific isolates. Also, the degree of fungal pathogenicity seemed to depend on the conidial viability of each isolate. Among the nine isolates tested, isolates INRS-242 and INRS-462 of *B. bassiana* and INRS-704 of *M. anisopliae* are those that have triggered the highest mortality in the eastern larch beetle populations tested. Other isolates of *B. bassiana* (INRS-453, INRS-200, INRS-452 and INRS-243) induced mortality ranging from 25% to 30%. Insects have shown a high infectious resistance when exposed to both isolates of *L. attenuatum* (INRS-1506 and INRS-1511). The highest conidial viability was obtained with isolate INRS-242 of *B. bassiana* (81.7%), followed by isolate INRS-704 of *M. anisopliae* (73.6%). Based on these results, the

isolate INRS-242 could be selected as a good candidate to develop a biocontrol strategy of *D. simplex* populations. It is an indigenous fungus of the Quebec ecoregion, and it was previously collected from cadavers of another Scolytinae species, the pine shoot beetle *Tomicus piniperda*, near Cookshire, QC, Canada. Our results with INRS-242 seem to support the hypothesis that isolates recovered from the same targeted host or a phylogenetically related species are generally more virulent than those from non-related species (Steinwender et al. 2010).

In Canada, the isolate *B. bassiana* INRS-242 has also demonstrated its effectiveness in infecting other Coleopteran species, such as the emerald ash borer, *Agrilus planipennis* (Johnny et al. 2012), the tarnished plant bug, *Lygus lineolaris*, the strawberry bud weevil, *Anthonomus signatus*, and the strawberry root weevil, *Otiorynchus ovatus* (Sabbahi et al. 2008a,b), the white pine weevil, *Pissodes strobi* (Trudel et al. 2007), the pine shoot beetle, *T. piniperda* (Lavallée et al. 2005), and the brown spruce longhorned beetle, *Tetropium fuscum* (Sweeney et al. 2005). However, the LC_{50} of *B. bassiana* INRS-242 seems to depend on the target insect species. For example, with a concentration of 7.4×10^8 conidia ml^{-1} , our results showed that *B. bassiana* INRS-242 killed 50% of *D. simplex* adults, whereas 10 times fewer conidia were required for other beetle species such as white pine weevil (Trudel et al. 2007) and pine shoot beetle (Lavallée et al. 2005). These variations in pathogenicity may be influenced by various factors, including cuticle structure (Bogus et al. 2010), insect defence mechanisms and the microbiome associated with the insect (Popa et al. 2012; Durand et al. 2015). Despite the fact that more investigations should be conducted, our results suggest that the INRS-242 isolate is a good candidate to be used in a biological control strategy against the eastern larch beetle populations.

The use of an assisted autodissemination device as a direct inoculation method is not a new approach, and different designs were adapted to control other insect pests (Dowd and Vega 2003; Liu and Bauer 2008). It was applied to control some Coleopteran species (Dowd and Vega 2003; Kreutz et al. 2004; Vega et al. 2007; Lyons et al. 2012). However, these systems used a dry conidia powder. For example, Kreutz and his collaborators (Kreutz et al. 2004) utilized the modified trap containing conidial powders of *B. bassiana* to inoculate european spruce bark beetle. As the results, they observed 84% of mortality in adults under the bark in which 78% presented mycosis. However, the persistence of entomopathogenic fungi in field environments is still a major challenge.

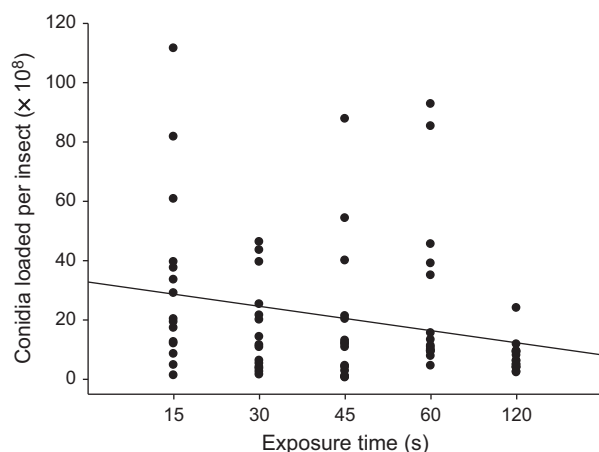


Fig. 4 Correlation between conidia loaded per adult and its exposure time on the pouch (Pearson test; $n = 75$; $r^2 = 0.08$; $P < 0.05$).

In the present study, dissemination chamber developed by Guertin and Lavallée was made from an inverted funnel fixed at the bottom part of a Lindgren trap containing a fungal-coated pouch. This structure protects the fungal-coated pouch from the rain and UV (Lyons et al. 2012). According to the same authors, conidial viability of INRS-242 isolate of *B. bassiana* in the dissemination chamber was higher than 74% after 43 days of outdoor exposure. Contrary to those observations, this isolate on foliar application lost their viability after only 6 days of direct sun exposure (Sabbahi et al. 2008a). Hence, the fungal-coated pouch can be used as effective dissemination source of the entomopathogenic fungus. Beetles walking on the fungal-coated pouch surface were successfully contaminated. Indeed, we observed that the *B. bassiana* INRS-242 isolate caused mortality reaching 98.0% of *D. simplex* adults with an average lethal dose of 2.1×10^9 conidia/insect. In comparison, if an immersion method had been used, then insects would have had to be treated with a fungal suspension at a concentration of 10^{12} conidia ml⁻¹ to obtain the same results (unpublished data). Our results demonstrated the efficiency of the fungus and the assisted fungal autodissemination device as an inoculation method.

In conclusion, our results demonstrate for the first time the development of a biological control strategy against *D. simplex* populations. Using an assisted fungal autodissemination device, the INRS-242 isolate of *B. bassiana* appears to be a promising agent to control *D. simplex*. Less damaging than chemical pesticides, the use of a selected natural insect pathogen is environmentally more acceptable. Being relatively specific, this strategy will target only the insect species attracted by the lures and the insect traps used with fewer negative side effects on non-targeted insect species. However, further studies are required to evaluate the effect of *B. bassiana* INRS-242 on biological reproduction of the eastern larch beetle as well as the effectiveness of the fungus and the device on field populations.

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References

- Abbott WS, 1925. A method of computing the effectiveness of an insecticide. *J Econ Entomol*, 18, 265–267.
- Baverstock J, Roy HE, Pell JK, 2010. Entomopathogenic fungi and insect behaviour: from unsuspecting hosts to targeted vectors. *Biol Control*, 55, 89–102.
- Behel RW, 2006. Importance of direct spray and spray residue contact for infection of *Trichoplusia ni* larvae by field application of *Beauveria bassiana*. *J Econ Entomol*, 99, 1120–1128.
- Benoit BJ, Blais JR, 1984. Pertes de bois causées par le dendroctone du mélèze. *Phytoprotection*, 65, 89.
- Bentz BJ, Régnière J, Fettig CJ, Hansen EM, Hayes JL, Hicke JA, Kelsey RG, Negron JF, Seybold SJ, 2010. Climate change and bark beetles of the Western United States and Canada: direct and indirect effects. *Bio-science*, 60, 602–613.
- Boguś MI, Czygier M, Golebiowski M, Kędra E, Kucińska J, Mazgajska J, Samborski J, Wieloch W, Włóka E, 2010. Effects of insect cuticular fatty acids on *in vitro* growth and pathogenicity of the entomopathogenic fungus *Conidiobolus coronatus*. *Exp Parasitol*, 125, 400–408.
- Butt TM, Ibrahim L, Ball BV, Clark SJ, 1994. Pathogenicity of the entomopathogenic fungi *Metarhizium anisopliae* and *Beauveria bassiana* against crucifer pests and the honey bees. *Biocontrol Sci Technol*, 4, 207–214.
- Dowd PF, Vega FE, 2003. Autodissemination of *Beauveria bassiana* by sap beetles (Coleoptera: Nitidulidae) to overwintering sites. *Biocontrol Sci Technol*, 13, 65–75.
- Durand A-A, Bergeron A, Constant P, Buffet JP, Déziel E, Guertin C, 2015. Surveying the endomicrobiome and ectomicrobiome of bark beetles: the case of *Dendroctonus simplex*. *Sci Rep*, 5, 17190.
- Finney DJ, 1971. Probit analysis, 3rd edn.. Cambridge University Press, Cambridge.
- Hajek AE, Huang B, Dubois T, Smith MT, Li Z, 2006. Field studies of control of *Anoplophora glabripennis* (Coleoptera: Cerambycidae) using fiber bands containing the entomopathogenic fungi *Metarhizium anisopliae* and *Beauveria brongniartii*. *Biocontrol Sci Technol*, 16, 329–343.
- Inglis GD, Enkerli J, Goettel MS, 2012. Laboratory techniques used for entomopathogenic fungi: Hypocreales. In: *Manual of techniques in invertebrate pathology*. Ed. by Lacey LA, Academic Press, San Diego, 189–253.
- Johny S, Kyei-Poku G, Gauthier D, Van Frankenhuyzen K, Krell PJ, 2012. Characterization and virulence of *Beauveria* spp. recovered from emerald ash borer in southwestern Ontario, Canada. *J Invertebr Pathol*, 111, 41–49.

- Kaya HK, Vega FE, 2012. Scope and basic principles of insect pathology. In: Insect pathology. Ed. by Vega FE, Kaya HK, Academic Press, San Diego, CA.
- Kocacevik S, Sevim A, Eroglu M, Demirbag Z, Demir I, 2015. Molecular characterization, virulence and horizontal transmission of *Beauveria pseudobassiana* from *Dendroctonus micans* (Kug.) (Coleoptera: Curculionidae). J Appl Entomol, 139, 381–389.
- Kocacevik S, Sevim A, Eroglu M, Demirbag Z, Demir I, 2016. Virulence and horizontal transmission of *Beauveria pseudobassiana* S.A. Rehner & Humber in *Ips sexdentatus* and *Ips typographus* (Coleoptera: Curculionidae). Turk J Agric For, 40, 241–248.
- Kreutz J, Zimmermann G, Vaupel O, 2004. Horizontal transmission of the entomopathogenic fungus *Beauveria bassiana* among the spruce bark beetle, *Ips typographus* (Col., Scolytidae) in the laboratory and under field conditions. Biocontrol Sci Technol, 14, 837–848.
- Langor DW, Raske AG, 1987a. Emergence, host attack, and overwintering behaviour of the Eastern Larch Beetle, *Dendroctonus simplex* LeConte (Coleoptera: Scolytidae) in Newfoundland. Can Entomol, 119, 975–983.
- Langor DW, Raske AG, 1987b. Reproduction and development of the eastern larch beetle, *Dendroctonus simplex* LeConte (Coleoptera: Scolytidae), in Newfoundland. Can Entomol, 119, 985–992.
- Lavallée R, Trudel R, Guertin C, Côté C, Coulombe C, Desroches P, De Groot P, Alfaro R, Kope H, Sweeny J, Thurston G, 2005. The use of *Beauveria bassiana* as a potential control method against the pine shoot beetle (*Tomicus piniperda*). IUFRO Meeting, Chapter S7.03.05, «Forest insect epidemics: Population dynamics, dispersal, and ecosystem impacts», University of Northern British Columbia, Canada, July 11–14, 12–13.
- Liu H, Bauer LS, 2006. Susceptibility of *Agrilus planipennis* (Coleoptera: Buprestidae) to *Beauveria bassiana* and *Metarhizium anisopliae*. J Econ Entomol, 99, 1096–1103.
- Liu H, Bauer LS, 2008. Microbial control of emerald ash borer, *Agrilus planipennis* (Coleoptera: Buprestidae) with *Beauveria bassiana* strain GHA: Greenhouse and field trials. Biol Control, 45, 124–132.
- Lyons DB, Lavallée R, Kyei-Poku G, Van Frankenhuyzen K, Johny S, Guertin C, Francese JA, Jones GC, Blais M, 2012. Towards the development of an autocontamination trap system to manage populations of emerald ash borer (Coleoptera: Buprestidae) with the native entomopathogenic fungus, *Beauveria bassiana*. J Econ Entomol, 105, 1929–1939.
- Maniania NK, Ouna E, Ahuya P, Frérot B, Félix A-E, Le Ru B, Calatayud P-A, 2011. Dissemination of entomopathogenic fungi using *Buseola fusca* male as vector. Biol Control, 58, 374–378.
- Manitoba Conservation Forestry Branch, 2014. Eastern larch beetle outbreak in Manitoba [WWW document]. <http://www.gov.mb.ca/conservation/forestry/health/eastern-larch.html>.
- Mckee FR, Aukema BH, 2015. Influence of temperature on the reproductive success, brood development and brood fitness of the eastern larch beetle *Dendroctonus simplex* LeConte. Agric For Entomol, 17, 102–112.
- Meyling NV, Pell JK, 2006. Detection and avoidance of an entomopathogenic fungus by a generalist insect predator. Ecol Entomol, 31, 162–171.
- Minnesota Department of Natural Resources - Division of Forestry, Forest Health Unit, 2014. Minnesota forest health report. St-Paul, Minnesota.
- Muratoglu H, Demirbag Z, Sezen K, 2011. The first investigation of the diversity of bacteria associated with *Leptinotarsa decemlineata* (Coleoptera: Chrysomelidae). Biologia, 66, 288–293.
- Ontario Ministry of Natural Resources, 2012. Forest Health Conditions in Ontario. Ontario, Canada.
- Pell JK, Macaulay EDM, Wilding N, 1993. A pheromone trap for dispersal of the pathogen *Zoophthora radicans* Brefeld. (Zygomycetes: Entomophthorales) amongst populations of the diamondback moth, *Plutella xylostella* L. (Lepidoptera: Yponomeutidae). Biocontrol Sci Technol, 3, 315–320.
- Popa V, Déziel E, Lavallée R, Baucé E, Guertin C, 2012. The complex symbiotic relationships of bark beetles with microorganisms: a potential practical approach for biological control in forestry. Pest Manag Sci, 68, 963–975.
- Sabbahi R, Merzouki A, Guertin C, 2008a. Efficacy of *Beauveria bassiana* (Bals.) Vuill. against the tarnished plant bug, *Lygus lineolaris* L., in strawberries. J Appl Entomol, 132, 124–134.
- Sabbahi R, Merzouki A, Guertin C, 2008b. Efficacy of *Beauveria bassiana* against the strawberry pests, *Lygus lineolaris*, *Anthonomus signatus* and *Otiiorhynchus ovatus*. J Appl Entomol, 132, 151–160.
- SAS, 2002. SAS System for window. SAS Institute, Cary, NC.
- Seybold SJ, Albers MA, Katovich SA, 2002. Eastern larch beetle. Forest insect and diseases Leaflet, 175, 10.
- Sezen K, Demir I, Demirbag Z, 2004. Study of the bacterial flora as a biological control agent of *Agelastica alni* L. (Coleoptera: Chrysomelidae). Biologia, 59, 327–331.
- Shimazu M, 2004. A novel technique to inoculate conidia of entomopathogenic fungi and its application for investigation of susceptibility of the Japanese pine sawyer, *Monochamus alternatus*, to *Beauveria bassiana*. Appl Entomol Zool, 39, 485–490.
- Srei N, 2011. Identification des facteurs permettant d'optimiser la production de *Beauveria bassiana* et les impacts sur la virulence. Master's Thesis. INRS-Institut Armand-frappier, 135.
- Steinwender BM, Krenn HW, Wegensteiner R, 2010. Different effects of the insect pathogenic fungus *Beauveria bassiana* (Deuteromycota) on the bark beetle *Ips*

- sexdentatus* (Coleoptera: Curculionidae) and on its predator *Thanasimus formicarius* (Coleoptera: Cleridae). *J Plant Dis Prot*, 117, 33–38.
- Sweeney J, Thurston G, Lavallée R, Trudel R, Desrochers P, Côté C, Guertin C, Todorova S, Kope HH, Alfaro R, 2005. *Beauveria bassiana* for control of the brown spruce longhorn beetle *Tetropium fuscum* (Fabr.) (Coleoptera: Cerambycidae). Proceedings, 16th U.S. Department of Agriculture interagency research forum on gypsy moth and other invasive species 2005, Loews Annapolis Hotel, Annapolis, Maryland, 2005 January 18–21, 82.
- Tanyeli E, Sevim A, Demirbag Z, Eroglu M, Demir I, 2010. Isolation and virulence of entomopathogenic fungi against the great spruce bark beetle, *Dendroctonus micans* (Kugelann) (Coleoptera: Scolytidae). *Biocontrol Sci Technol*, 20, 695–701.
- Trudel R, Lavallée R, Guertin C, Côté C, Todorova SI, Alfaro R, Kope H, 2007. Potential of *Beauveria bassiana* (Hyphomycetes: Moniliales) for controlling the white pine weevil, *Pissodes strobi* (Col., Curculionidae). *J Appl Entomol*, 131, 90–97.
- Vega FE, Dowd PF, Bartelt RJ, 1995. Dissemination of microbial agents using an autoinoculating device and several insect species as vectors. *Biol Control*, 5, 545–552.
- Vega FE, Dowd PF, Lacey LA, Pell JK, Jackson DM, Klein MG, 2007. Dissemination of beneficial microbial agents by insects. In: *Fields manual of techniques in invertebrate pathology*. Ed. by Lacey LA, Kaya HK, Springer, Dordrecht, 127–146.
- Wegensteiner R, 2004. Pathogens in bark beetles. In: *Bark and wood boring insects in living tree in Europe, a synthesis*. Ed. by Lieutier F, Day KR, Battisti A, Gregoire J-C, Evans HF, Kluwer Academic Publisher, Dordrecht, 291–313.
- Wegensteiner R, Wermelinger B, Herrmann M, 2015. Natural enemies of bark beetles: predators, parasitoids, pathogens, and nematodes. In: *Bark beetles: biology and ecology of native and invasive species*. Ed. by Vega FE, Hofstetter RW, Academic Press, San Diego, 247–304.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. Hosts and geographic origins of entomopathogenic fungal isolates used in the experiment.